

AE 06: Joining country populations with continents

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Goal

Our ultimate goal in this application exercise is to create a bar plot of total populations of continents, where the input data are:

1. Countries and populations
2. Countries and continents

```
library(tidyverse) # for data wrangling and visualization
library(scales)    # for pretty axis breaks
```

Data

Countries and populations

These data come from [The World Bank](#) and reflect population counts as of 2022.

```
population <- read_csv("data/world-pop-2022.csv")
```

Let's take a look at the data.

```
population
```

```
# A tibble: 217 x 3
  country      year population
  <chr>      <dbl>      <dbl>
1 Afghanistan 2022    41129.
2 Albania     2022     2778.
```

```

3 Algeria                2022    44903.
4 American Samoa         2022      44.3
5 Andorra                2022      79.8
6 Angola                 2022   35589.
7 Antigua and Barbuda    2022     93.8
8 Argentina              2022  46235.
9 Armenia                2022   2780.
10 Aruba                 2022    106.
# i 207 more rows

```

Continents

These data come from [Our World in Data](#).

```
continents <- read_csv("data/continents.csv")
```

Let's take a look at the data.

```
continents
```

```

# A tibble: 285 x 4
  entity                code    year continent
  <chr>                 <chr>  <dbl> <chr>
1 Abkhazia             OWID_ABK  2015 Asia
2 Afghanistan          AFG      2015 Asia
3 Akrotiri and Dhekelia OWID_AKD  2015 Asia
4 Aland Islands        ALA      2015 Europe
5 Albania              ALB      2015 Europe
6 Algeria              DZA      2015 Africa
7 American Samoa       ASM      2015 Oceania
8 Andorra              AND      2015 Europe
9 Angola               AGO      2015 Africa
10 Anguilla             AIA      2015 North America
# i 275 more rows

```

Exercises

- **Think out loud:**

- Which variable(s) will we use to join the `population` and `continents` data frames?

country and entity

- We want to create a new data frame that keeps all rows and columns from `population` and brings in the corresponding information from `continents`. Which join function should we use?

`left_join()`

- **Demo:** Join the two data frames and name assign the joined data frame to a new data frame `population_continents`.

```
population_continents <- population %>%  
  left_join(continents, join_by(country == entity))
```

- **Demo:** Take a look at the newly created `population_continents` data frame. There are some countries that were not in `continents`. First, identify which countries these are (they will have NA values for `continent`).

```
print(population_continents)
```

```
# A tibble: 217 x 6
```

	country <chr>	year.x <dbl>	population <dbl>	code <chr>	year.y <dbl>	continent <chr>
1	Afghanistan	2022	41129.	AFG	2015	Asia
2	Albania	2022	2778.	ALB	2015	Europe
3	Algeria	2022	44903.	DZA	2015	Africa
4	American Samoa	2022	44.3	ASM	2015	Oceania
5	Andorra	2022	79.8	AND	2015	Europe
6	Angola	2022	35589.	AGO	2015	Africa
7	Antigua and Barbuda	2022	93.8	ATG	2015	North America
8	Argentina	2022	46235.	ARG	2015	South America
9	Armenia	2022	2780.	ARM	2015	Asia
10	Aruba	2022	106.	ABW	2015	North America

```
# i 207 more rows
```

- **Demo:** All of these countries are actually in the `continents` data frame, but under different names. So, let's clean that data first by updating the country names in the `population` data frame in a way they will match the `continents` data frame, and then joining them, using a `case_when()` statement in `mutate()`. At the end, check that all countries now have continent information.

```
# add code here
```

- **Think out loud:** Which continent do you think has the highest population? Which do you think has the second highest? The lowest?

Add your response here.

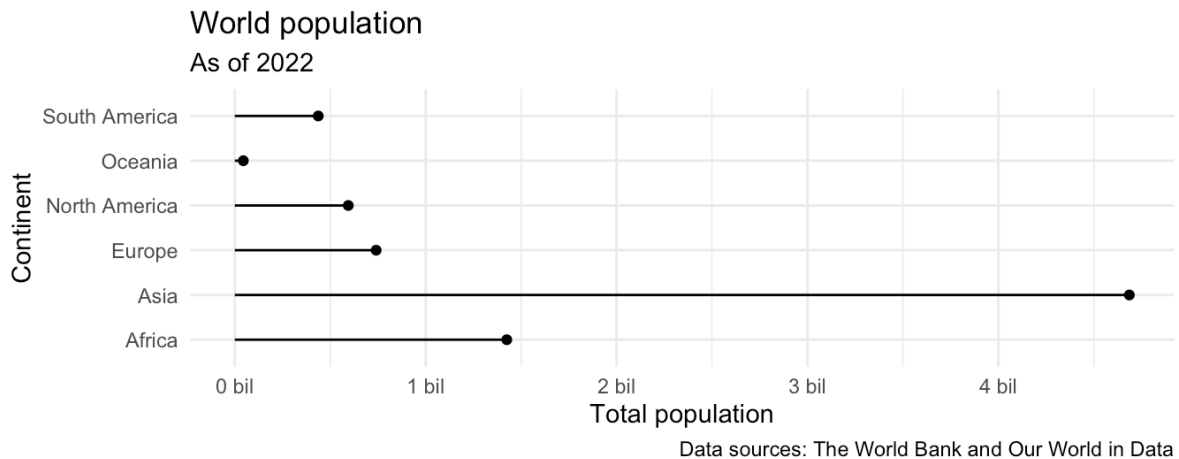
- **Demo:** Create a new data frame called `population_summary` that contains a row for each continent and a column for the total population for that continent, in descending order of population. Note that the function for calculating totals in R is `sum()`.

```
# add code here
```

- **Your turn:** Make a bar plot with total population on the y-axis and continent on the x-axis, where the height of each bar represents the total population in that continent.

```
# add code here
```

- **Your turn:** Recreate the following plot, which is commonly referred to as a **lollipop plot**. Hint: Start with the points, then try adding the `segments`, then add axis labels and `caption`, and finally, as a stretch goal, update the x scale (which will require a function we haven't introduced in lectures or labs yet!).



```
# add code here
```

- **Think out loud:** What additional improvements would you like to make to this plot.

Add your response here.